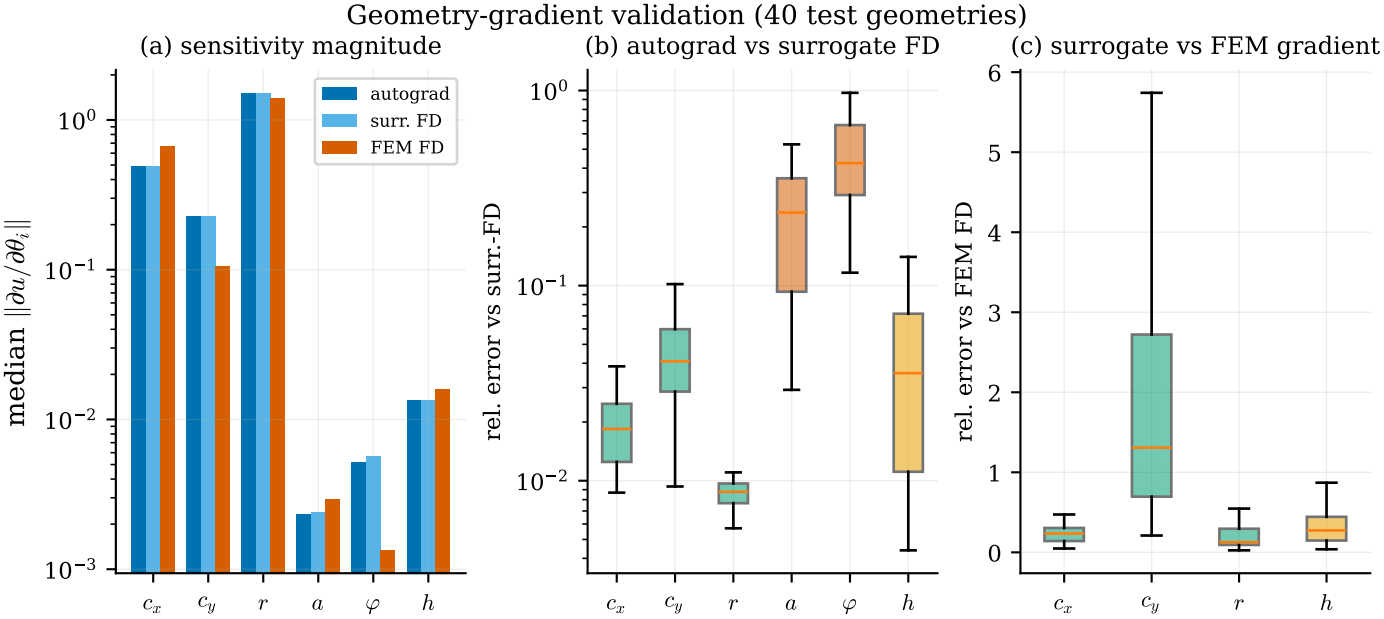
Geometry-gradient validation (40 test geometries)
(a) sensitivity magnitude
(b) autograd vs surrogate FD
(c) surrogate vs FEM gradient
autograd
surr. FD
FEM FD
median $\|\partial u/\partial \theta_i\|$
rel. error vs surr.-FD
rel. error vs FEM FD
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
1
2
3
4
5
6
c_x
c_y
r
a
φ
h